

Task 5: Exploratory Data Analysis (EDA)

Titanic Dataset

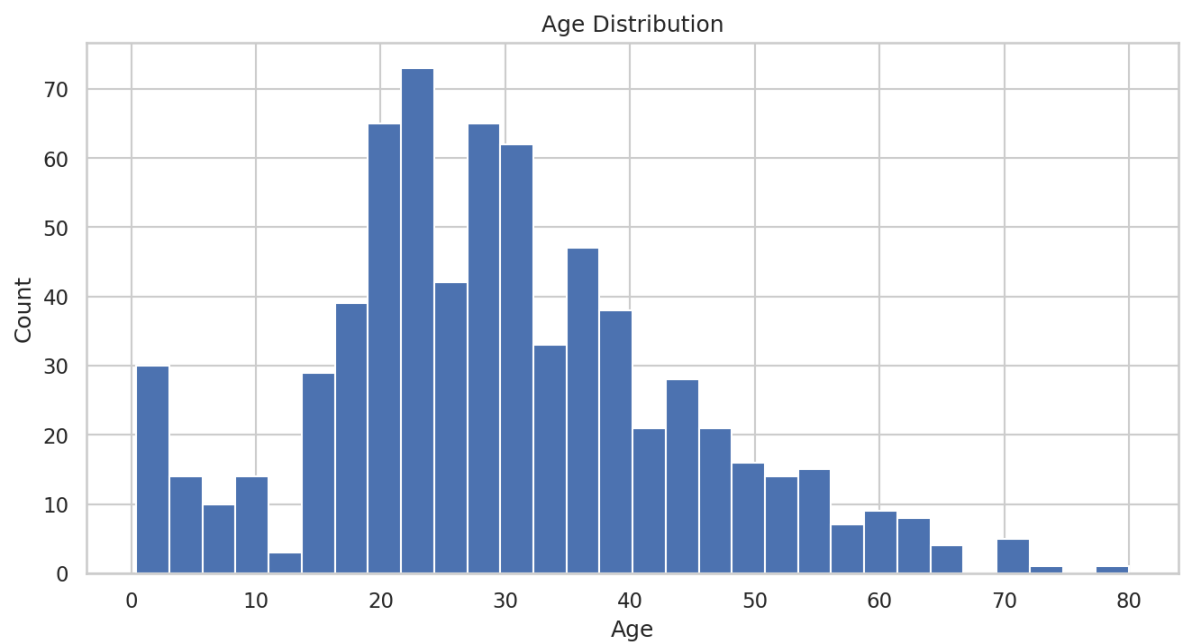
Objective: Extract insights using visual and statistical exploration with Pandas, Matplotlib and Seaborn.

1. Dataset Overview

The dataset contains **891** passenger records and **12** variables.

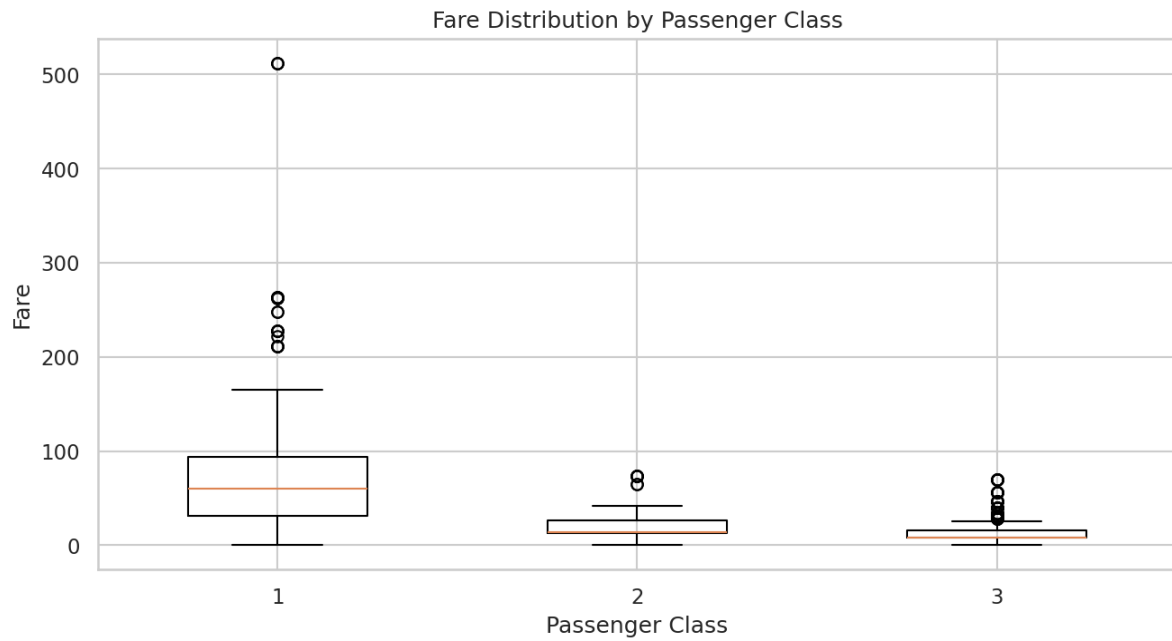
Metric	Value
Rows	891
Columns	12
Overall survival	38.38%
Missing Cabin	687 (77.1%)
Missing Age	177 (19.9%)
Missing Embarked	2 (0.2%)

2. Age Distribution



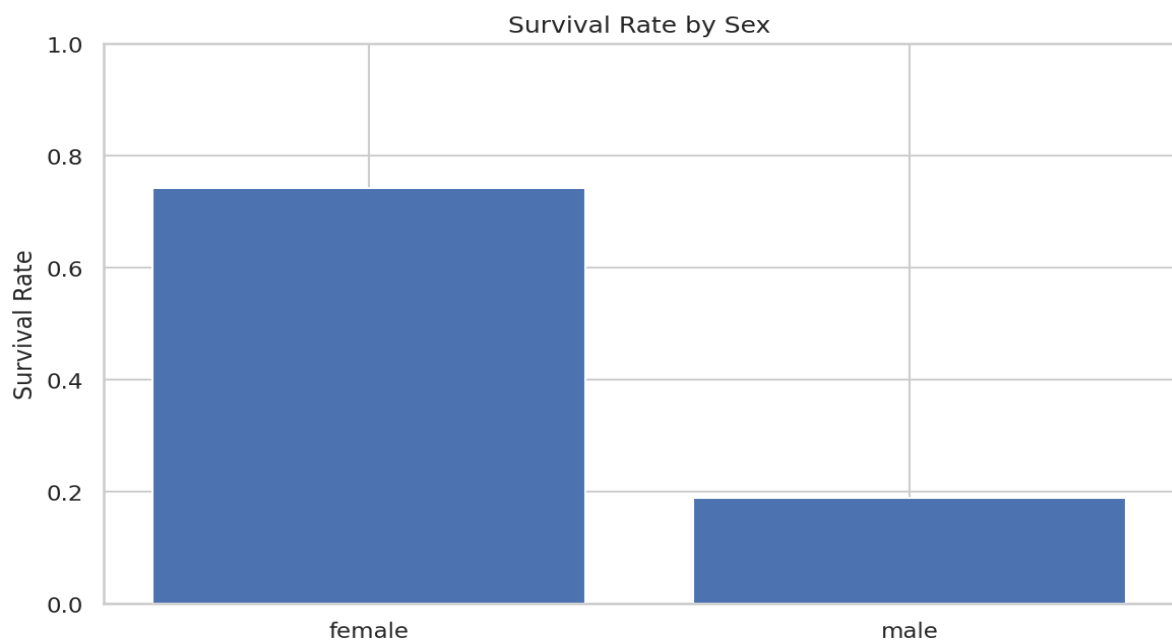
Observation: Most passengers were young to middle-aged adults.

3. Fare and Class Analysis



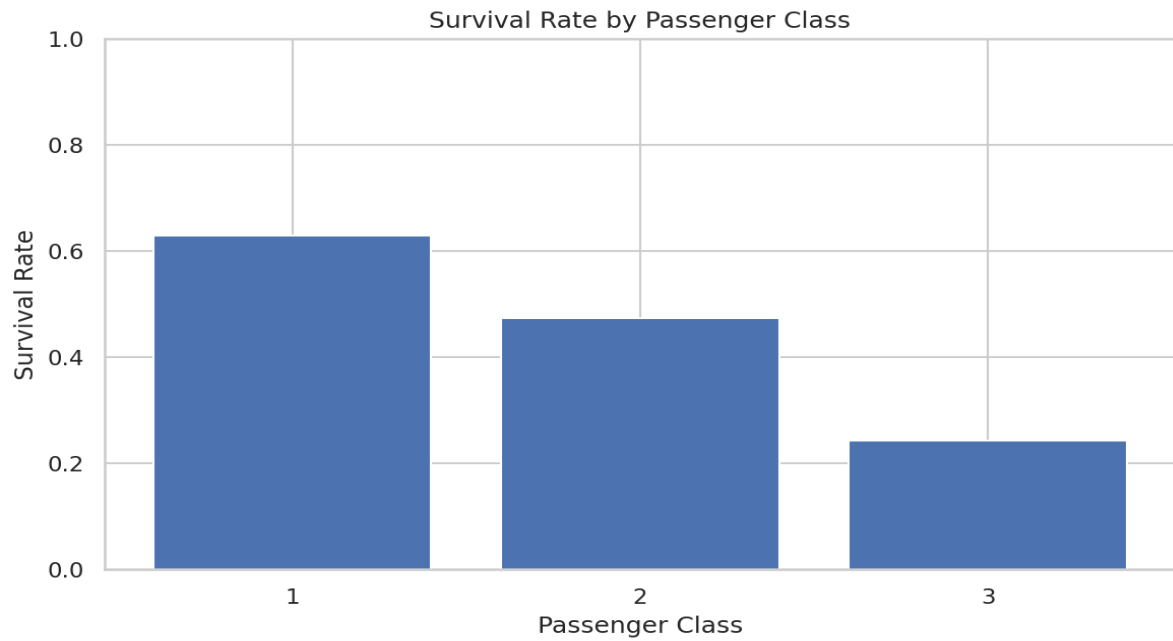
Observation: First-class fares are substantially higher and contain high-value outliers.

4. Survival by Sex



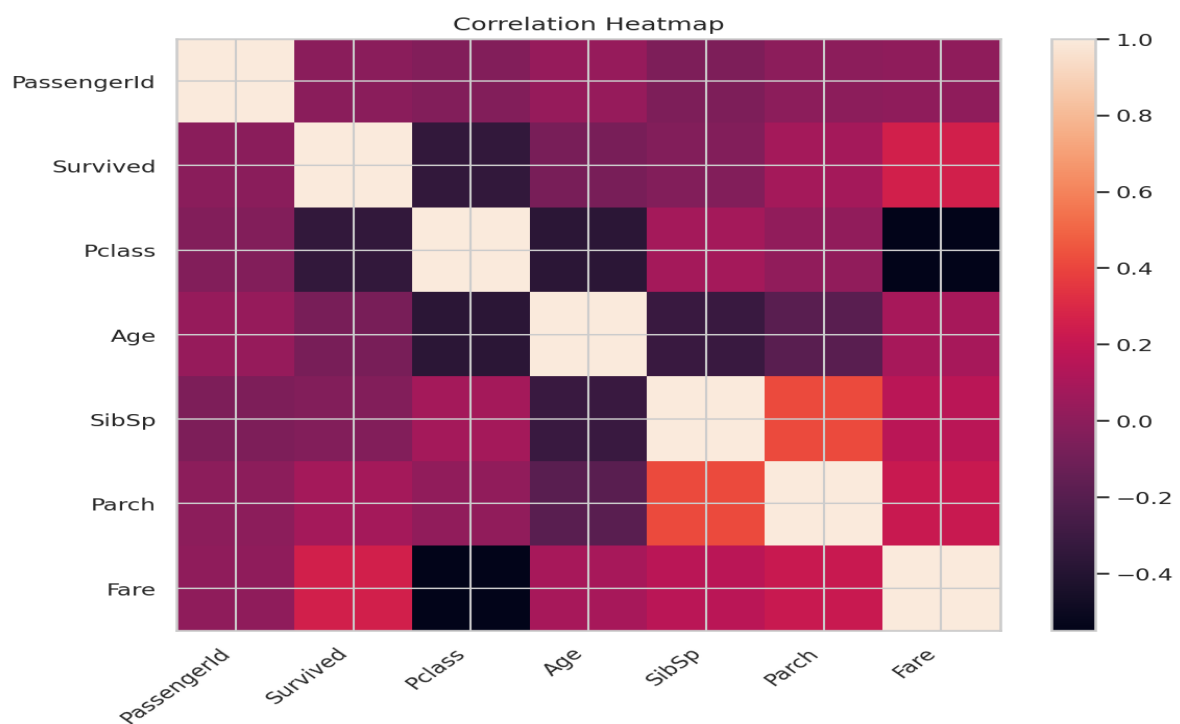
Observation: Female passengers had a substantially higher survival rate.

5. Survival by Passenger Class



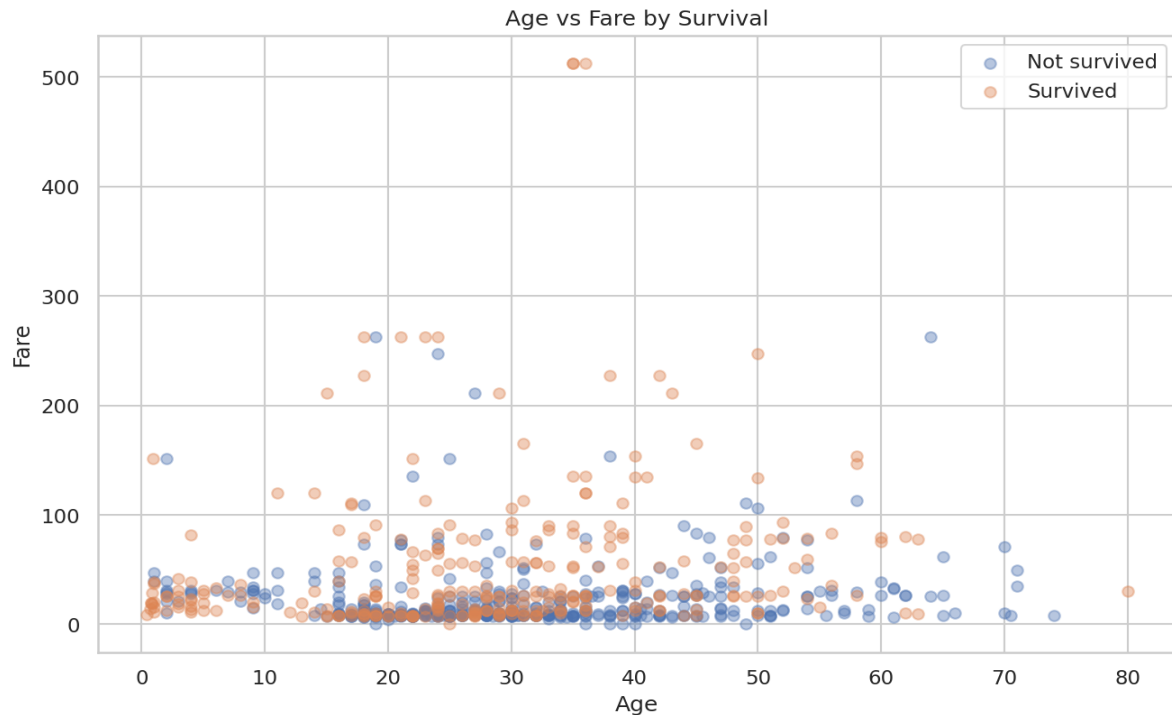
Observation: Survival decreases from first class to third class.

6. Correlation Analysis



Observation: Class and fare are strongly related, while survival is associated with class and fare.

7. Scatterplot Analysis



Observation: Age and fare do not show a simple linear relationship.

8. Key Findings

- Overall survival was approximately 38%.
- Female passengers survived at a much higher rate than male passengers.
- First-class passengers had the highest survival rate and third-class passengers the lowest.
- Fare contains noticeable high-value outliers.
- Missing values are concentrated in Cabin, Age and Embarked.
- Age alone does not clearly separate survivors from non-survivors.
- Sex and passenger class are the clearest patterns associated with survival.

9. Conclusion

The Titanic EDA demonstrates how descriptive statistics and visualization reveal patterns before predictive modeling. Sex and passenger class emerge as especially important variables associated with survival, while missing values and fare outliers require attention during data preparation.